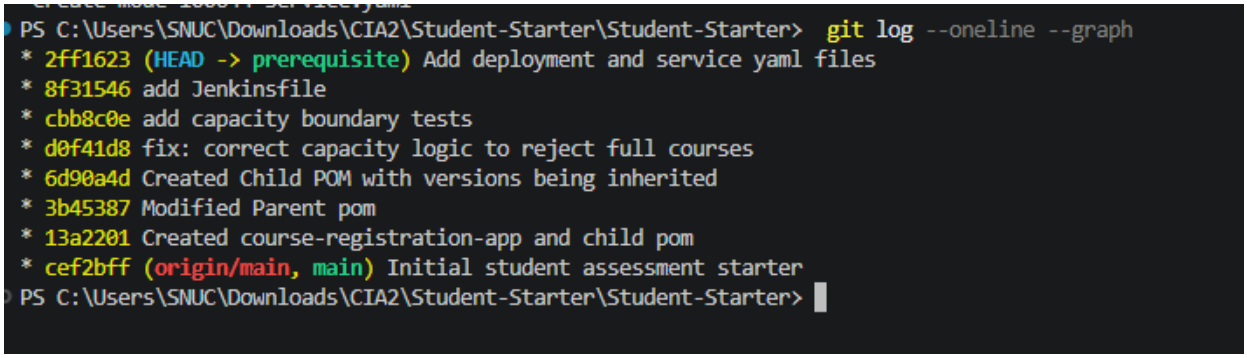
DevOps Lab CIA 2 Screenshots

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1. Git Evidence



2. Maven Configuration

Build Configuration (parent pom.xml)

The parent POM establishes common dependency management, shared properties, and standard plugin configurations across sub-modules.<?xml version="1.0" encoding="UTF-8"?>

<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
https://maven.apache.org/xsd/maven-4.0.0.xsd">

<modelVersion>4.0.0</modelVersion>

<groupId>com.university</groupId>

<artifactId>course-registration</artifactId>

<version>1.0-SNAPSHOT</version>

```
<packaging>pom</packaging>

<modules>

    <module>course-registration-app</module>

</modules>

<properties>

    <maven.compiler.source>17</maven.compiler.source>

    <maven.compiler.target>17</maven.compiler.target>

    <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>

    <java.version>17</java.version>

    <junit.version>5.11.0</junit.version>

</properties>

<dependencyManagement>

    <dependencies>

        <dependency>

            <groupId>org.junit.jupiter</groupId>

            <artifactId>junit-jupiter</artifactId>

            <version>${junit.version}</version>

            <scope>test</scope>

        </dependency>
```

```
    </dependencies>
</dependencyManagement>
<build>
    <pluginManagement>
        <plugins>
            <plugin>
                <groupId>org.apache.maven.plugins</groupId>
                <artifactId>maven-compiler-plugin</artifactId>
                <version>3.13.0</version>
                <configuration>
                    <source>${java.version}</source>
                    <target>${java.version}</target>
                </configuration>
            </plugin>
            <plugin>
                <groupId>org.apache.maven.plugins</groupId>
                <artifactId>maven-surefire-plugin</artifactId>
                <version>3.5.0</version>
            </plugin>
```

```
        </plugins>
    </pluginManagement>
  </build>
</project>
```

Build Configuration (child pom.xml)

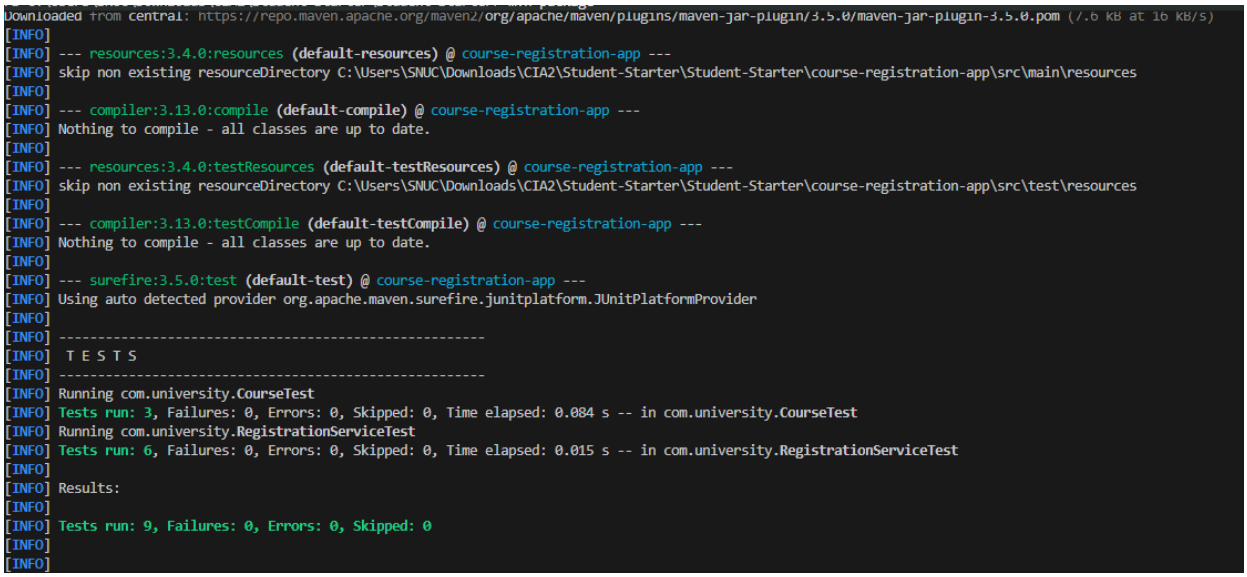
The child module inherits properties and dependency versions from the parent model to ensure version alignment across the project.<?xml version="1.0" encoding="UTF-8"?>

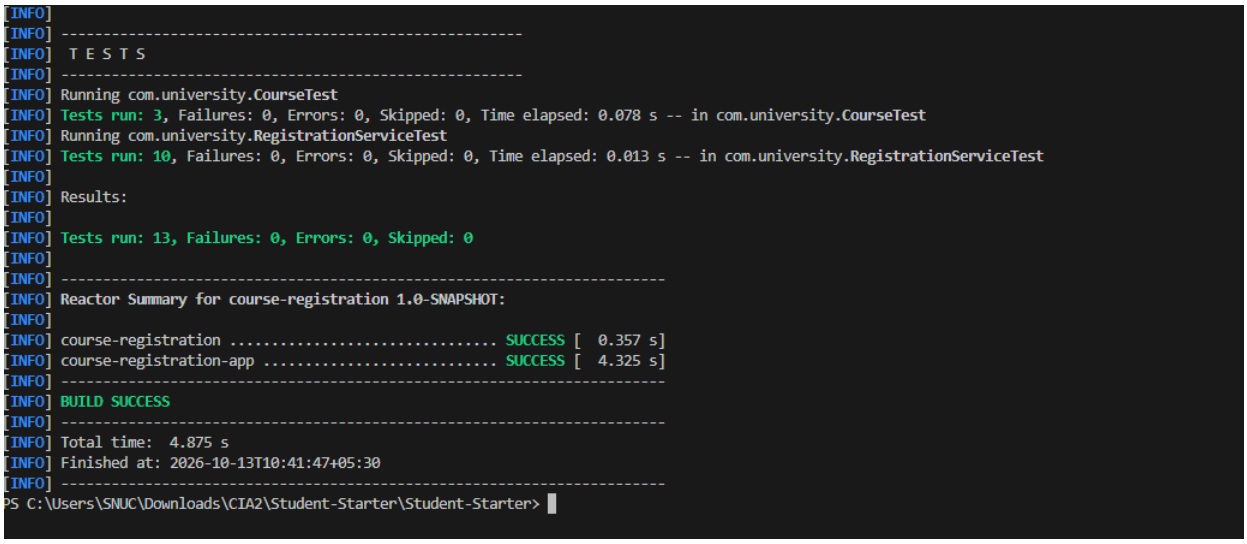
```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
https://maven.apache.org/xsd/maven-4.0.0.xsd">
  <modelVersion>4.0.0</modelVersion>
  <parent>
    <groupId>com.university</groupId>
    <artifactId>course-registration</artifactId>
    <version>1.0-SNAPSHOT</version>
  </parent>
  <artifactId>course-registration-app</artifactId>
  <packaging>jar</packaging>
  <dependencies>
    <dependency>
```

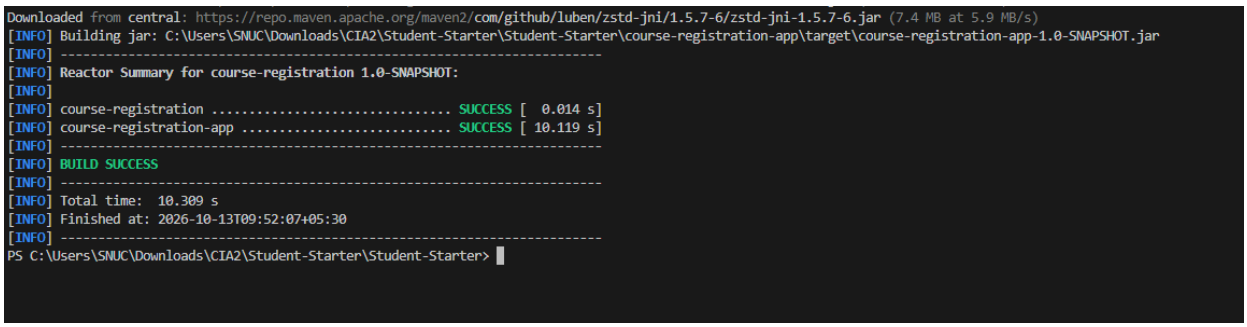
```
        <groupId>org.junit.jupiter</groupId>
        <artifactId>junit-jupiter</artifactId>
        <scope>test</scope>
    </dependency>
</dependencies>
<build>
    <plugins>
        <plugin>
            <groupId>org.apache.maven.plugins</groupId>
            <artifactId>maven-compiler-plugin</artifactId>
            <configuration>
                <source>${java.version}</source>
                <target>${java.version}</target>
            </configuration>
        </plugin>
        <plugin>
            <groupId>org.apache.maven.plugins</groupId>
            <artifactId>maven-surefire-plugin</artifactId>
        </plugin>
```

```
        </plugins>
    </build>
</project>
```

Build and Test Execution







3. Jenkins Pipeline Automation

Jenkinsfile

```
pipeline {

    agent any

    stages {

        stage('Compile') {

            steps {

                sh 'mvn clean compile'

            }

        }

        stage('Test') {

            steps {

                sh 'mvn test'

            }

            post {

                always {

                    junit 'course-registration-app/target/surefire-reports/TEST-*.xml'

                }

            }

        }
```

```
        stage('Package') {
            steps {
                sh 'mvn package -DskipTests'
            }
            post {
                success {
                    archiveArtifacts artifacts: 'course-registration-app/target/*.jar', fingerprint: true
                }
            }
        }
    }
}
```

4. Kubernetes Deployment & Self-Healing Evidence

Manifest Specifications

Application Deployment (deployment.yaml)

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: demo-app-deployment
```

```
  labels:
    app: demo-app
spec:
  replicas: 3
  selector:
    matchLabels:
      app: demo-app
  template:
    metadata:
      labels:
        app: demo-app
    spec:
      containers:
        - name: demo-app-container
          image: demo-app:1.0.0
          imagePullPolicy: IfNotPresent
          ports:
            - containerPort: 8080
          resources:
```

```
          requests:
            memory: "128Mi"
            cpu: "100m"
          limits:
            memory: "256Mi"
            cpu: "250m"
        livenessProbe:
          httpGet:
            path: /actuator/health
            port: 8080
          initialDelaySeconds: 15
          periodSeconds: 10
        readinessProbe:
          httpGet:
            path: /actuator/health
            port: 8080
          initialDelaySeconds: 10
          periodSeconds: 5
```

NodePort Service (service.yaml)

```
apiVersion: v1
```

```
kind: Service
metadata:
  name: demo-app-service
  labels:
    app: demo-app
spec:
  type: NodePort
  selector:
    app: demo-app
  ports:
    - port: 80
      targetPort: 8080
      nodePort: 30080
      protocol: TCP
```

```
PS C:\Users\SNUC\Downloads\CIA2\Student-Starter\Student-Starter> kubectl get pods
>>
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-75cc477c85-ld7hn   1/1     Running   0          106s
nginx-deployment-75cc477c85-lrdtq   1/1     Running   0          9s
nginx-deployment-75cc477c85-vqj6w   1/1     Running   0          106s
PS C:\Users\SNUC\Downloads\CIA2\Student-Starter\Student-Starter>
```

```
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-75cc477c85-8rjdc   1/1     Running   0          12s
nginx-deployment-75cc477c85-ld7hn   1/1     Running   0          22s
nginx-deployment-75cc477c85-vqj6w   1/1     Running   0          22s
PS C:\Users\SNUC\Downloads\CIA2\Student-Starter\Student-Starter> $POD_NAME = (kubectl get pods -l app=nginx -o jsonpath='{.items[0].metadata.name}')
PS C:\Users\SNUC\Downloads\CIA2\Student-Starter\Student-Starter> kubectl delete pod $POD_NAME
pod "nginx-deployment-75cc477c85-8rjdc" deleted from default namespace
PS C:\Users\SNUC\Downloads\CIA2\Student-Starter\Student-Starter> kubectl get pods
>>
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-75cc477c85-ld7hn   1/1     Running   0          106s
nginx-deployment-75cc477c85-lrdtq   1/1     Running   0          9s
nginx-deployment-75cc477c85-vqj6w   1/1     Running   0          106s
PS C:\Users\SNUC\Downloads\CIA2\Student-Starter\Student-Starter>
```